

Ablation Study on Model Components

Configuration	Components			Metrics			
	Decomp.	Routing	Sampling	MAE↓	RMSE↓	$\delta_{<1.25}\uparrow$	Δ MAE
Full Model	✓	✓	✓	0.285	1.362	0.841	—
w/o Signal Decomp.	—	✓	✓	0.341	1.524	0.793	+19.6%
w/o Dual-Mask Routing	✓	—	✓	0.318	1.467	0.812	+11.6%
w/o Balanced Sampling	✓	✓	—	0.306	1.421	0.823	+7.4%
Baseline (w/o all)	—	—	—	0.427	1.893	0.712	+49.8%

Δ MAE shows relative degradation from the full model. Best results in **bold**.

Ablated Variants